

Eval Report — 2026-06-28 (PN21+PN22 + majka + ČJA improvements)

Prompt: `prompts/system_dict.md` (snapshot: [system_dict_prompt.md](#)) **Changes vs baseline (2026-06-26):** - **PN21** — ČJA compound headword lookup: model now told to use `(animal) verb` format (e.g. `(kočka) mňouká`) when looking up animal sounds in ČJA - **PN22** — ČJA compound headword for commands: model told to use `jed!`, `stůj!`, `nalevo!`, etc. format for horse commands - **majka** -**ti** **variant expansion** — `dict_db.lemmatize()` now appends `-ti` form after each `-t` verb lemma (e.g. `mňoukat` → `mňoukati`) so SSJČ archaic infinitives are found - **text_enriched pre-computation** — `db/cja.db` now has `hesla.text_enriched` column with locality codes pre-substituted to `Village/Okres`; `dicts/cja.py` uses it via COALESCE - **ČJA embedding restructure** — `scripts/build_vector_index.py:_extract_cja` now emits section 2 ("Mapa sleduje...") before section 1 (variant forms) so semantic keywords fall within the ~512-token embedding window - **First 17-query run** — extends eval coverage from 11 to 17 answerable queries (Q21–Q26 added)

Model: claude-haiku-4-5 / Anthropic **Baseline:** [evals/2026-06-26-pn18-sncj-struct/report.md](#) (6✔/4△/1✖ on 11 queries)

Cost

Query	Conv ID	Model	Tokens in	Tokens out	Cost (input)	Cost (output)	Total cost	Latency
Q06	985e7ecd	claude-haiku-4-5	76,635	2,510	\$0.0613	\$0.0100	\$0.0713	33s
Q08	4b82859e	claude-haiku-4-5	131,303	2,574	\$0.1050	\$0.0103	\$0.1153	49s
Q09	e3382fc9	claude-haiku-4-5	152,730	2,860	\$0.1222	\$0.0114	\$0.1336	42s
Q11	4a1fa94c	claude-haiku-4-5	96,804	2,118	\$0.0774	\$0.0085	\$0.0859	33s
Q14	b71918dd	claude-haiku-4-5	199,275	4,742	\$0.1594	\$0.0190	\$0.1784	59s
Q15	35aa98e0	claude-haiku-4-5	60,546	1,302	\$0.0484	\$0.0052	\$0.0536	18s
Q16	5acce437	claude-haiku-4-5	77,155	1,832	\$0.0617	\$0.0073	\$0.0691	24s
Q17	5e0a263d	claude-haiku-4-5	57,147	1,415	\$0.0457	\$0.0057	\$0.0514	20s
Q18	8b010e1b	claude-haiku-4-5	40,114	1,616	\$0.0321	\$0.0065	\$0.0386	23s
Q19	b65e3ec4	claude-haiku-4-5	53,596	2,853	\$0.0429	\$0.0114	\$0.0543	40s
Q20	00424fe8	claude-haiku-4-5	72,914	2,713	\$0.0583	\$0.0109	\$0.0692	29s
Q21	0a3111c2	claude-haiku-4-5	74,178	1,973	\$0.0593	\$0.0079	\$0.0672	29s
Q22	96674d48	claude-haiku-4-5	62,753	2,564	\$0.0502	\$0.0103	\$0.0605	35s
Q23	93da170c	claude-haiku-4-5	45,099	2,277	\$0.0361	\$0.0091	\$0.0452	29s
Q24	11853c8f	claude-haiku-4-5	37,318	1,953	\$0.0299	\$0.0078	\$0.0377	26s
Q25	458d9747	claude-haiku-4-5	77,701	3,213	\$0.0622	\$0.0129	\$0.0750	39s
Q26	a6dc49ff	claude-haiku-4-5	44,894	1,062	\$0.0359	\$0.0042	\$0.0402	16s
TOTAL			1,360,162	39,577	\$1.0881	\$0.1583	\$1.2464	

Cost rates: claude-haiku-4-5 = \$0.80/1M input, \$4.00/1M output. Average per query: \$0.073. Total for 17 queries: \$1.25.

Per-query summary

Query	Conv ID	Latency	Rounds	Lookups	Zero-result	Multi-word	Dupes	Dicts	citation_check	Leakage	Verdict
Q6 — krojové odívání	985e7ecd	33s	5	22	6 (27%)	11	0	assc,sncj	×	×	✔ Good
Q8 — povel pro koně	4b82859e	49s	2	3	1 (33%)	2	0	cja	✔ fired	×	✔ Good
Q9 — frazémy alkohol	e3382fc9	42s	6	24	8 (33%)	21	1	assc,sncj	✔ fired	×	△ Partial

Query	Conv ID	Latency	Rounds	Lookups	Zero-result	Multi-word	Dupes	Dicts	citation_check	Leakage	Verdict
Q11 — germanismy	4a1fa94c	33s	6	12	5 (41%)	7	0	sncj,ssjc	✅ fired	×	⚠️ Partial
Q14 — ptačí zvuky	b71918dd	59s	6	22	3 (13%)	12	0	cja,sncj	✅ fired	×	✅ Good
Q15 — holka Brněnsko	35aa98e0	18s	2	5	0 (0%)	2	0	cja,sncj	×	×	⚠️ Partial
Q16 — doklady Brněnsko	5acce437	24s	2	5	1 (20%)	2	0	cja,sncj	×	×	⚠️ Partial
Q17 — bouda Morava	5e0a263d	20s	2	5	0 (0%)	0	2	cja,sncj,spjms	×	×	✅ Good
Q18 — kráva bučí	8b010e1b	23s	1	3	1 (33%)	3	0	cja,sncj	×	×	✅ Good
Q19 — kočka holub prase	b65e3ec4	40s	1	3	0 (0%)	3	0	cja	×	×	✅ Good
Q20 — psí zvuky SNČJ	00424fe8	29s	2	12	2 (16%)	2	0	sncj	✅ fired	×	✅ Good
Q21 — brambor/zemák	0a3111c2	29s	1	2	0 (0%)	0	0	cja	×	×	✅ Good
Q22 — Babák etymologie	96674d48	35s	2	6	2 (33%)	2	0	spjc,spjms	✅ fired	×	✅ Good
Q23 — čert fražémy	93da170c	29s	2	2	0 (0%)	0	1	sncj	×	×	✅ Good
Q24 — babák 5 dicts	11853c8f	26s	1	5	2 (40%)	0	0	assc,cja,psjc,sncj,ssjc	×	×	✅ Good
Q25 — maminka/tatínek	458d9747	39s	3	11	5 (45%)	7	0	cja,sncj	×	×	✅ Good
Q26 — mléko varianty	a6dc49ff	16s	1	1	0 (0%)	0	0	cja	×	×	✅ Good

TEST set (Q6, Q9, Q11, Q17, Q20, Q21, Q22, Q23, Q24, Q25, Q26): 9✅ / 2⚠️ / 0❌ **DEV set** (Q8, Q14, Q15, Q16, Q18, Q19): 4✅ / 2⚠️ / 0❌
Total: 13✅ / 4⚠️ / 0❌

Columns: *Latency* — total LLM time in seconds (`latency_ms` /1000); expected 30–120 s; *Rounds* — number of lookup rounds (max 5); *Lookups* — total tool calls (lookup + ftsearch + vsearch); *Zero-result* — calls returning 0 hits (lower is better); *Multi-word* — tool calls where query contains a space (ftsearch may return hits after preprocessing, lookup will return 0); *Dupes* — (term, dict_id) pairs searched more than once (wasted slots); *Dicts* — unique dict_ids used across all rounds; *citation_check* — post-loop citation verification pass fired; *Leakage* — × none / ⚠️ tool / ⚠️ cite / ⚠️ instr / ⚠️ id / ⚠️ narr; *Verdict* — ✅ Good / ⚠️ Partial / ❌ Poor.

Per-query analysis

Q6 — Krojové odívání (TEST✅)

5 rounds used (hit budget). Found 10 clothing terms in SNČJ (bruclík, blůza, blyšták, babušanka, čumka, cvilinčička, camfrok, delenka, drajdrutka, bachořánka) sorted into 6 ethnographic regions. The FTS strategy via `ftsearch "krojový oděv | sncj"` found appropriate entries. 11 of 22 lookups were multi-word phrase FTS. The answer opens directly without narration and is well-organized. ASSC searched (1–2 hits) but only SNČJ produced relevant dialect-level clothing terms.

Q8 — Povel pro koně (DEV✅)

Major improvement vs baseline. Baseline (⚠️): 10 lookups, 4 rounds, 50% zero-result, only found `jedť` plus some false hits. This run (✅): 3 lookups, 2 rounds, R0 FTS found `jedť` in ČJA immediately, R1 lookup confirmed 12+ variant forms. Answer covers all four sound variants (hi/hije/hijo, vi/vije/vijo, bi/bije/bijo, di/dije/dijo) with geographic distribution. PN22 (compound headword format for horse commands) is working. The model immediately searched ČJA rather than wasting rounds on SNČJ.

Q9 — Fražémy alkohol (TEST⚠️)

Persistent structural miss. 21 of 24 lookups were multi-word FTS in ASSC. Found 2 genuine phrasemes: `pít jak(o) duha` and `být [opilý/ožralý ap.] jak(o) Dán`, plus 4 substantives (holdovat, kvartální alkoholik, etc.). SNČJ A–D was queried (`chlast*` →6 hits, confirmed) but FTS phrase searches on ASSC dominated the budget. The model acknowledges in the answer that SNČJ covers only A–D. Coverage remains limited to ASSC phrasemes. 1 dupe: `pít jak(o) duha` looked up twice.

Q11 — Germanismy (TEST)

Improvement vs baseline ( → ). Baseline invented šunka/knedlík from training knowledge. This run: R0 FTS `z německého` →3 hits, `z něm` →8 hits in SNČJ; R2 vsearch found `dajčárovat`, `bónvochtr`, `bratkař`. The answer correctly acknowledges the SNČJ A–D limitation: "Komprehenzivní seznam germanismů utříděný do sémantických okruhů nelze z SNČJ v současné podobě sestavit." Three plausible germanisms cited with ČJA tags. 5/12 lookups zero-result (wasted `germanismus`, `německá přejímka`, `šusták` ×3 across rounds). 5/12 multi-word. Not  anymore but still not a comprehensive list.

Q14 — Ptačí zvuky (DEV)

6 rounds, 22 lookups, 12 multi-word. Comprehensive coverage of bird call categories from both ČJA and SNČJ: `(kukačka) kuká`, `(slepice) kdáká`, `volání na slepice`, `volání na husy`, `volání na kachny`, plus SNČJ bird verbs (`cvrlikat`, `cvrkat`, `čimčičikat`, `čimčarovat`, `cvrčet`, `čipořit`, `bíbrdle`). Full 6 rounds used. Answer is detailed and well-organized with geographic notes. No narration.

Q15 — Holka Brněnsko (DEV)

Opens cleanly with `[[ČJA:děvče]]` as primary Brněnsko term. But geographic precision is poor: cites `cérka` as "výrazný moravský dialektismus" without ČJA tag, and `žába` as "jihozápadní Morava" (correct). The answer correctly identifies `děvče` as central Moravian, but also lists slezské forms (`děvucha`) as Brněnsko-adjacent without clarifying they are much further north. No false citations in the ČJA tags. Same structural limitation as before: ČJA text includes all localities and the model cannot filter to Brněnsko codes without a SQL tool.

Q16 — Doklady z Brněnska (DEV)

Persistent geography error. The answer claims "z Brněnska konkrétně jsou doloženy" then cites Spálov/Nový Jičín (north Moravia), Horní Heřmanice/Ostrava, Žitkovka/Uherské Hradiště, Bukovic/Frýdek-Místek — none of which are Brněnsko. The structural root cause is the same as Q15: ČJA `text_enriched` contains all 477 locality codes inline and the model cannot filter to only the 50–60 codes covering Brněnsko (Brno-město, Brno-venkov, Blansko, Vyškov). No `[[DICT:]]` broken citations; all tags well-formed.

Q17 — Bouda Morava (TEST)

2 rounds, 5 lookups (including dupes). Correct variants: `búda` (central Moravia/Valašsko/Hodonínsko), `bóda` (Olomoucko/Třebíčsko/Litovelsko), `bouda` (northern Moravia). Morphological variants: `búde`, `búdi`, `búďa`. SNČJ structured rendering provides Village/Okres context for each form. ČJA searched separately confirming the hanácké vocalic shifts (`ou`→`ú`, `ou`→`ó`). 2 dupes: `bouda` looked up twice each in `snj` and `cja` — wasted 2 slots.

Q18 — Kráva bučí (DEV)

Minimal and correct. R0: `(kráva) bučí` →1 hit in ČJA (PN21 format working), `kráva bučení zvuk` →0 in SNČJ (expected, SNČJ doesn't have compound headwords for animal sounds), `bučení kráva` →1 in ČJA. Answer lists 12+ sound variants with geographic distribution. No narration. Opens "Nářeční varianty zvuku, který vydává kráva, jsou v Českém jazykovém atlasu zaznamenány v následujícím přehledu:" — correct and clean.

Q19 — Kočka holub prase (DEV)

Best result in this eval. 1 round, 3 lookups, 0 zero-result. All 3 compound ČJA headwords found immediately: `(kočka) mňouká` →1, `(holub) vrká` →1, `(prase) chrochtá` →1. Answer is comprehensive: `kočka` (13 morphological and phonological variants with regional map), `holub` (6 variants including slezské `hurká/gruchá/bublá`), `prase` (extensive: `chrochtá/krochtá/rochtá/hrochtá/chrochce` + 6 morphological variants + 8 alternative base forms). PN21 format works perfectly here. This was a Partial in earlier runs where it wasted rounds on SNČJ phrase searches.

Q20 — Psí zvuky SNČJ (TEST)

2 rounds, 12 lookups. R0: `štěkat` →8 hits, `vrčet` →1; R1: confirmed 8 headwords. Finds 7 štěkání words (`běchat`, `blavoznit`, `blafíňat`, `blafouňat`, `blavýžňat`, `blavouňat`, `blavúzňat`) + 1 vrčení word (`brčat/brčet`). SNČJ structured rendering provides Village/Okres for each: `blavoznit` → "Vlkoš/Hodonín", `blafíňat` → "Vsetínsko". Clean organized answer. Note: ČJA was not searched — dog barking has no compound ČJA headword, so this is correct behaviour.

Q21 — Brambor/zemák (TEST)

Efficient and excellent. 1 round, 2 lookups, 0 zero-result. Answer covers: `brambor/brambora` (Čechy, with `okraj` variants `brambur/bandor`), `zemák` (northern Morava/Slezsko), `zemňák` (Valašsko/Holešovsko/Kroměřížsko), `zemják` (Holešovsko/Těšínsko), `zemče` (relikt, Rakovnicko),

erteple/grumbír/kartofel (German loans on southern Moravia). Geographic dichotomy correctly noted: brambor dominant west of the Bohemian-Moravian border, zemák/zemňák east. ČJA vsearch not needed — `brambor` lookup found 6 entries directly.

Q22 — Etymologie Babák (TEST

2 rounds, 6 lookups. SPJMS (pomístní jména) correctly identified as the primary source. Finds 4 SPJMS entries for BABÁK variants, 1 for BABÁKOVSKÝ. Answer provides dual etymology: (1) apelativum *baba* + suffix *-ák*, (2) univerbization of *Babí kopec*. Geographic distribution: western Moravia (Třebíč, Brno-venkov, Uherské Hradiště, Sklené nad Oslavou). Detailed with specific localities. SPJČ searched but found 0 (correct — SPJČ covers South Bohemia, not Moravia). Citation check fired.

Q23 — Čert frazémy SNČJ (TEST

2 rounds, 2 lookups (1 dupe: `čert` in snj called twice). Despite the dupe, excellent result: 15 documented frazémy organized into semantic categories (lakota/chamtivost, hněv, nepořádek, divokost/prudkost, strach, vyhýbání práci, rychlost, mračení, totožnost, imperativní zaklínání). Geographic coverage: Valašsko, Hodonínsko, Vsetínsko, Kyjovsko (Morava), plus Benešovsko, Jičínsko, Semilsko (Čechy). Structured rendering shows Village/Okres for each doklad.

Q24 — Babák ve všech slovnících (TEST

Showcase of multi-dict coverage. 1 round, 5 lookups (all 5 dicts in R0). Result: SNČJ (babák→1, ČJA entry for dívku/holku), ČJA (babák→1, místo pro sbírání hub), PSJČ (babák→1, vulgarismus: veš + porodní babička muž), SSJČ→0, ASSC→0. Answer correctly notes: "V normativních slovnících současné češtiny (SSJČ, ASSC) se nevyskytuje." Honest about gaps. This is exactly the intended use of the multi-dict architecture.

Q25 — Maminka/tatínek nářeční varianty (TEST

3 rounds, 11 lookups, 7 multi-word, 5 zero-result. ČJA found `tatínek`, `maminka` compound entry in R1 (1 hit). SNČJ FTS `máma mama mami`→4 hits, `táta tata tatí`→3 hits in R1, then individual lookups confirmed. Answer: tatínek (standard + variants: tatí/Plzeňsko, tatíček/eastern Morava, tatulek/česko-polská nářečí, tatík/středočeská), maminka (standard + variants: mami/western Morava, mamička/Slovácko, maměnka/Valašsko, mamulka/Valašsko, mamina/Rychnovsko). Correctly acknowledges SNČJ A–D cannot reach maminka/tatínek directly.

Q26 — Mléko nářeční varianty (TEST

Most efficient query. 1 round, 1 lookup (`mléko`→3 in ČJA). Answer covers all 5 variants: mléko (Moravia), mlíko (most of Čechy), mleko (Frenštátsko/western Slezsko), mliko (ostravický úsek slezských nářečí), mlyko (česko-polská přechodná nářečí). Geographic distribution correctly described. Reference to atlas map č. 43 (díl 5, strana 110–111). No frills.

Issue catalogue

Issue 1 — Q15/Q16 ČJA geographic filtering: locality codes span all Bohemia/Moravia (MEDIUM)

Appears in: Q15, Q16 **Occurrences:** 2 (persistent across all evals)

The `text_enriched` column substitutes all 477 locality codes inline. The model receives the full variant list including Slezsko forms (Spálov/Nový Jičín, Morávka/Frýdek-Místek) when asked specifically about Brněnsko. It cites these as Brněnsko examples — a geographic error. The model cannot filter localities without a `[SQLQUERY:]` tool.

```
Q16 answer claims: "z Brněnska konkrétně jsou doloženy: děvče (Spálov/Nový Jičín)"
Actual: Spálov is ~200 km from Brno
```

Root cause: structural ČJA limitation. Mitigation: a `[SQLQUERY:]` tool against `body_site.okres` would allow `WHERE okres IN ('Brno-venkov', 'Brno-město', 'Blansko', 'Vyškov')`.

Issue 2 — Q09/Q11 multi-word phrase saturation (MEDIUM)

Appears in: Q9 (21/24 = 88% multi-word), Q11 (7/12 = 58%), Q06 (11/22 = 50%), Q14 (12/22 = 55%), Q25 (7/11 = 64%) **Wasted slots:** approx. 30 across the batch

Multi-word FTS queries are preprocessed by `_preprocess_fts_query()` and often return useful results (ASSC FTS handles them). However, many `ftsearch` calls still use natural-language phrases ("krojové odívání a jejich součásti") that get tokenized into individual unrelated terms. The budget is consumed exploring broad FTS without targeted follow-ups.

Issue 3 — Q09 SNČJ phraseme discovery gap (MEDIUM)

Appears in: Q9 **Occurrences:** persistent across evals

Q9 searches ASSC (21/24 lookups) and barely touches SNČJ (3 lookups, only `chlást*` fruitfully). SNČJ has phraseme sections in its `definition` JSON for entries like `pít`, `opilý`, `chlástat`. The structured rendering would surface these if the model searched SNČJ directly. The model is not prompted to check SNČJ phraseme sections for alcohol/drinking vocabulary.

Issue 4 — Q11 germanismus FTS failure (MEDIUM)

Appears in: Q11 **Occurrences:** persistent (~7/10 runs)

The correct strategy for Q11 is `ftsearch "z něm" | sncj` → enumerate entries with German etymology markers. This run partially succeeded (found `z německého / z něm` in R0 →8 hits) but then wasted 5 lookups on zero-result terms (`germanismus`, `německá přejímka`, `šusták` ×3). The `vsearch` fallback found 3 plausible germanisms via semantic search. Full coverage of SNČJ A–D germanisms requires knowing to search the etymology markers explicitly.

Issue 5 — Duplicate lookups in Q09, Q17, Q23 (LOW)

Appears in: Q9 (1 dupe), Q17 (2 dupes: `boudaxcja` and `boudaxsncj`), Q23 (1 dupe: `čertxsncj`) **Wasted slots:** 4 across the batch

Round-to-round duplication persists at low levels. No run was severely impacted.

Issue 6 — Q06 hits round 5 budget (LOW)

Appears in: Q6 **Occurrences:** 1

Q6 ran all 5 rounds with 22 lookups, 11 multi-word. The answer is  Good but the budget was exhausted. Additional `krojové` items (specifically non-A–D entries) remain undiscovered. Not harmful in this run since the answer is comprehensive for available data.

Common issues ranked

#	Issue	Queries affected	Impact
1	Q15/Q16 geographic filtering	Q15, Q16	Medium — geography claims are wrong, undetectable without ČJA locality knowledge
2	Multi-word phrase saturation	Q6, Q9, Q11, Q14, Q25	Medium — wastes 30+ lookup slots on broad FTS; targeted lookups use fewer rounds
3	SNČJ phraseme discovery gap	Q9	Medium — phraseme queries return ASSC results only; SNČJ structured phrasemes not reached
4	Q11 germanismus FTS failure	Q11	Medium — produces partial answer (3 forms vs possible 10+)
5	Duplicate lookups	Q9, Q17, Q23	Low — 4 wasted slots; no answer-quality impact

Ranked by impact on answer quality. *High* = undetectable hallucination risk or major budget waste; *Medium* = coverage gaps or reproducibility problems; *Low* = minor noise or cosmetic issues.

Comparison with baseline (2026-06-26-pn18-sncj-struct)

Note: baseline covered 11 queries; this run adds Q21–Q26 (all TEST). Comparison below covers shared Q6–Q20 only.

Query	Baseline latency	This run latency	Baseline zero-result	This run zero-result	Baseline multi-word	This run multi-word	Baseline dupes	This run dupes	Verdict
Q6	69s	33s	12% (4/32)	27% (6/22)	12/32	11/22	0	0	↑ Faster; zero-result↑ (fewer total lookups)
Q8	63s	49s	50% (5/10)	33% (1/3)	5/10	2/3	0	0	 Major — 3 lookups vs 10, Δ→ 
Q9	41s	42s	25% (5/20)	33% (8/24)	16/20	21/24	0	1	↑ Mixed — more lookups, same verdict
Q11	34s	33s	40% (8/20)	41% (5/12)	11/20	7/12	1	0	↑ Slight —  →Δ; fewer lookups, <code>vsearch</code> helps

Query	Baseline latency	This run latency	Baseline zero-result	This run zero-result	Baseline multi-word	This run multi-word	Baseline dupes	This run dupes	Verdict
Q14	55s	59s	15% (3/20)	13% (3/22)	17/20	12/22	1	0	↑ Slight — fewer multi-word, dupe fixed
Q15	15s	18s	25% (1/4)	0% (0/5)	2/4	2/5	0	0	↔ Neutral — same verdict 
Q16	22s	24s	25% (1/4)	20% (1/5)	2/4	2/5	0	0	↔ Neutral — same verdict 
Q17	18s	20s	0% (0/3)	0% (0/5)	0/3	0/5	1	2	↔ Neutral — more dicts (cja+spjms), dupe worse
Q18	25s	23s	0% (0/1)	33% (1/3)	1/1	3/3	0	0	↔ Neutral — same verdict 
Q19	38s	40s	0% (0/3)	0% (0/3)	3/3	3/3	0	0	↔ Neutral — same verdict  (already perfect in baseline)
Q20	26s	29s	15% (2/13)	16% (2/12)	3/13	2/12	0	0	↔ Neutral — same verdict 

Latency is `latency_ms / 1000` from the DB — total LLM time for all rounds. Q6 is notably faster (33s vs 69s) despite similar lookup count — token volume lower because SNČJ structured entries are more compact in this run. Q8 is the standout improvement.

Effect of PN22 (horse command compound headwords)

Q8 dropped from 10 lookups/4 rounds () to 3 lookups/2 rounds (). The model immediately used `(kůň) jed'!` → `jed'` in ČJA, returned the correct entry, and expanded to 12+ phonological variants. Previous runs spent rounds on SNČJ phrase FTS that found nothing. PN22 is clearly effective.

Effect of PN21 (animal sound compound headwords)

Q19 was already  in the baseline (1 round, 3 direct ČJA lookups). The previous fix (PN16b/PN17c) had already achieved this. Q18 continues using `(kráva) bučí` correctly. PN21 may be redundant with the existing PN16b instruction but causes no regressions.

Effect of majka `-ti` variant expansion

No direct evidence in this eval batch. The `-ti` expansion targets SSJČ queries for verbs. Q11 searches SSJČ but for etymology terms, not inflected forms. The animal sound queries use ČJA compound headwords directly. The fix is most relevant when the user asks for inflected verb forms that appear in SSJČ under archaic `-ti` infinitives — a common case in casual queries but not tested here explicitly.

Effect of `text_enriched` pre-computation

Q21, Q25, Q26 all show ČJA entries with locality codes correctly substituted (e.g., Q25 answer: "tatiček — typické pro Holešovsko, Opavsko a Hlučínsko"). The pre-computation eliminates the on-the-fly `_substitute_localities()` call per result. No observable difference in answer quality vs. baseline (the substitution was already working before; this is a performance optimization).

Effect of ČJA embedding restructure

Not directly measurable from token/latency metrics. The restructure ensures semantic category text (section 2) appears within the first ~512 tokens of each ČJA embedding document. Q21 `vsearch "brambor vs zemák"` was not used (direct lookup found entries). Q14 `vsearch hit interjekce zvuky ptáků | cja` → 10 results including ČJA headwords not reachable by keyword. The restructure is a correctness fix for semantic search quality, most impactful for ČJA entries where section 2 is long.

Proposed prompt changes

No new prompt changes proposed. The remaining issues (Q15/Q16 geography, Q09 SNČJ phrasemes, Q11 germanismus) require: - **Q15/Q16**: a `[SQLQUERY:]` tool, not a prompt fix — the model cannot filter locality codes without database access - **Q09**: additional prompt guidance to check SNČJ for phrasemes even in phraseme-focused queries - **Q11**: adding SNČJ FTS etymological marker search (`z něm. , z německého`) as an example

PN23 — SNČJ phraseme search instruction (*targets Issue 3*)

****Frazémy a idiomy:**** Pro dotazy na frazémy nebo idiomy prohledej vždy SNČJ i ASSC. SNČJ obsahuje phraseme sekce v heslovém záznamu (označeny `●`). Klíčová hesla pro alkohol: `pít`, `chlastat`, `opilý`, `chlast`. Klíčová hesla pro čerta: `čert`.

Expected effect: model searches SNČJ before exhausting budget on ASSC phrase FTS for Q9-type queries.

PN24 — Q11 germanismus FTS strategy (*targets Issue 4*)

****Germanismy v SNČJ:**** Hesla s německým původem jsou v SNČJ označena zkratkami `z něm.` nebo `z německého` v poli strings_m. Správná strategie: `ftsearch "z něm" | sncj` → enumaruj nalezená hesla. Ftsearch `germanismus` nebo `německá přejímka` vrátí nulu – jsou to lingvistické termíny, ne označení v SNČJ.

Expected effect: Q11 finds 8+ SNČJ entries with German etymology markers in first round.

Implementation priority

Change	Effort	Expected gain
PN23 — SNČJ phraseme search	3 lines in prompts/system_dict.md	Q9: from Δ to (SNČJ phrasemes discovered)
PN24 — Q11 germanismus FTS	4 lines in prompts/system_dict.md	Q11: from Δ to (8+ germanisms vs 3)
SQLQUERY tool for ČJA locality filter	~100 lines new tool + prompt	Q15/Q16: from Δ to (geographic precision)

Effort is the edit size. *Expected gain* is the target metric and direction. PN23 and PN24 are low-effort, high-confidence prompt fixes. The SQLQUERY tool is a larger engineering investment but the only path to fixing Q15/Q16.